Code No: 136BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2024
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	How to use a design pattern? Write the guidelines.	[2]
b)	Give design patterns used in Smalltalk MVC.	[3]
c)	What is embellished object structure?	[2]
d)	What is the structure of document?	[3]
e)	Write the intent of Abstract Factory Pattern?	[2]
f)	What is Composite Pattern?	[3]
g)	What are the consequences of Decorator Pattern?	[2]
h)	Give the intent of Façade Pattern.	[3]
i)	What is the motivation of Interpreter Pattern?	[2]
j)	Write the structure of Command Pattern.	[3]

PART – B

(50 Marks)

2.a) List and explain different ways of organizing design patterns.
b) Give the step-by-step approach to apply a design pattern effectively. [5+5]

OR

3.a) How does design patterns solve design problems?
b) Give an overview of organizing a catalog. [5+5]

4.a) How a Design pattern solves the design problem? Illustrate with an example.
b) Explain in detail about Supporting Multiple Look-and-Feel Standards. [5+5]

OR

5.a) Describe transparent enclosure in embellishing the user interface.
b) Elaborate on User Operations, Spelling Checking and Hyphenation design problem. [5+5]

6.a) What are consequences of Builder pattern? Give the implementation of Builder pattern.
b) Discuss in detail about creational pattern. [5+5]

OR

7.a) What are the benefits of Singleton pattern? Discuss the implementation issues of singleton pattern in detail.
b) In detail discuss the Motivation, Structure, Collaborations and implementation of the Prototype Pattern. [5+5]

8.a) The Façade pattern and the Adapter pattern may seem similar? What is the essential difference between the two? Explain.

b) Explain in brief about Bridge pattern. [5+5]

OR

9. What are two classic examples of decorators? How does the Decorator pattern help to decompose the problem? [10]

10.a) Briefly discuss about Memento pattern.

b) Discuss the Participants, Collaborations, Sample Code, and Related Patterns of Command Pattern and Interpreter Pattern. [5+5]

OR

11.a) Give the Motivation, Participants, Structure and Implementation of Strategy Pattern and Visitor Pattern.

b) Differentiate Mediator pattern and the Template Method pattern. [5+5]

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